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T81-558: Applications of Deep Neural Networks

Module 13: Advanced/Other Topics

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- For more information visit the class website.

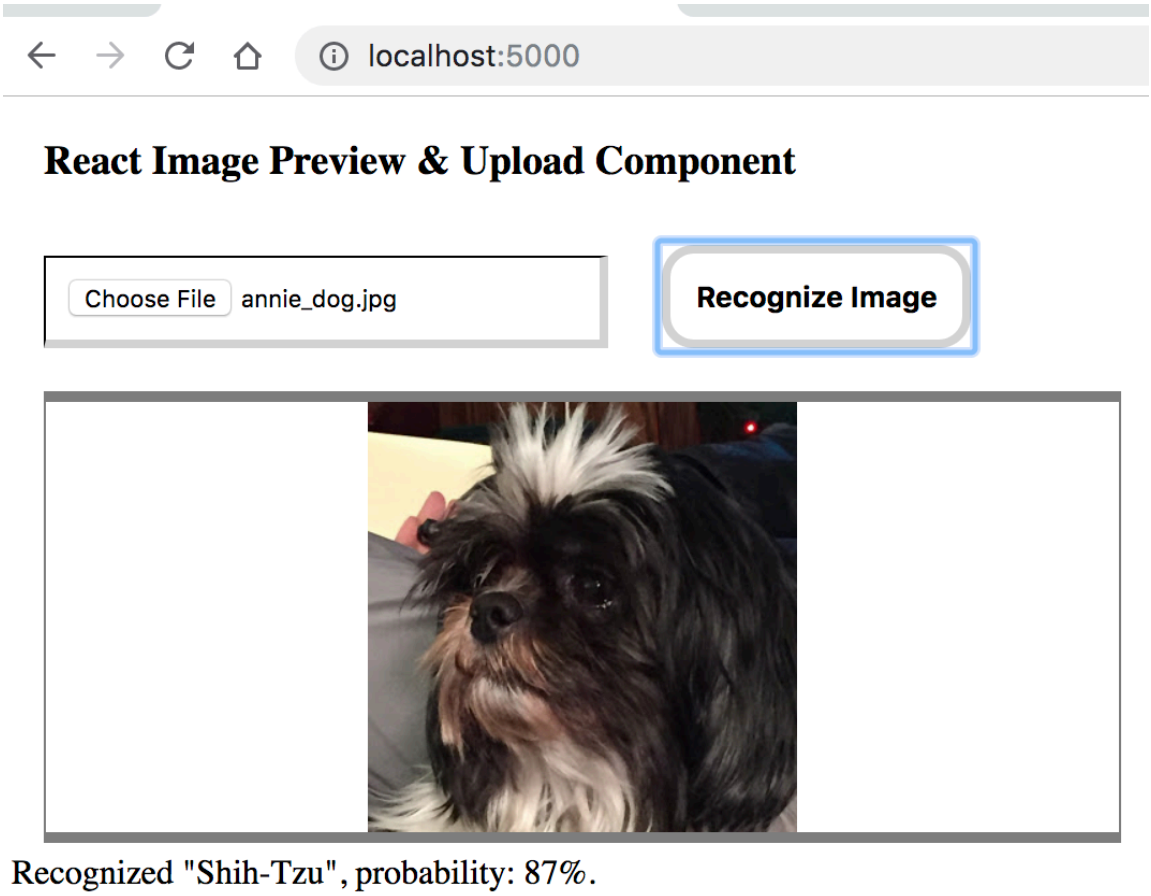
Module 13 Video Material

- Part 13.1: Flask and Deep Learning Web Services [Video] [Notebook]
- Part 13.2: Interrupting and Continuing Training [Video] [Notebook]
- **Part 13.3: Using a Keras Deep Neural Network with a Web Application** [Video] [Notebook]
- Part 13.4: When to Retrain Your Neural Network [Video] [Notebook]
- Part 13.5: Tensor Processing Units (TPUs) [Video] [Notebook]

Part 13.3: Using a Keras Deep Neural Network with a Web Application

In this part, we will extend the image API developed in Part 13.1 to work with a web application. This technique allows you to use a simple website to upload/predict images, such as in Figure 13.WEB.

Figure 13.WEB: AI Web Application



I added neural network functionality to a simple ReactJS image upload and preview example. To do this, we will use the same API developed in Module 13.1. However, we will now add a ReactJS website around it. This single-page web application allows you to upload images for classification by the neural network. If you would like to read more about ReactJS and image uploading, you can refer to the blog post that provided some inspiration for this example.

I built this example from the following components:

- GitHub Location for Web App
- image_web_server_1.py - The code both to start Flask and serve the HTML/JavaScript/CSS needed to provide the web interface.
- Directory WWW - Contains web assets.
  - index.html - The main page for the web application.
  - style.css - The stylesheet for the web application.
  - script.js - The JavaScript code for the web application.

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